Jack He (Zhe He)

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Education

University of California, Los Angeles (UCLA)
M.S. in Computer Science — Expected Dec 2026
B.S. in Computer Science; Double Major in Applied Mathematics — Jun 2025
GPA: 3.92/4.0 (undergraduate) | Dean's Honors List, 2021–2025
Relevant Coursework: Efficient Deep Learning, Advanced Deep Learning, Machine Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Algorithms, Linear Algebra, Probability & Statistics

Technical Skills

Languages: C++, C, Python, SQL, Shell, Java, JavaScript
ML & Inference: PyTorch, TensorFlow, NumPy, scikit-learn, llama.cpp, low-bit quantization, LoRA fine-tuning
Kernel Optimization: Linear attention, prefill/decode, matmul/GEMM, FX64 numerical functions
Systems & Tools: Boost.Asio, Redis, PostgreSQL, Git, Docker, Google Cloud Platform, Google Cloud Build

Experience

Quadric — Jun 2026 – Present
AI Kernel Engineer Intern
- Implemented **Qwen3.6 linear-attention kernels** from scratch for prefill and decode on the Chimera **GPNPU**, supporting production inference alongside MoE blocks.
- Optimized linear-attention **prefill/decode** and hardware-level routing, achieving a **3.5x decode speedup**.
- Optimized **matmul and GEMM kernels** for the GPNPU, achieving a **7.6x matmul efficiency improvement**.
- Improved **FX64 square-root and division** implementations, resolving precision-scaling issues to improve numerical accuracy in edge cases.

UCLA, Bolei Zhou Lab — Mar 2024 – Mar 2026
Research Assistant, Embodied AI — *Advisor: Prof. Bolei Zhou*
- Developed a **vision-language-action (VLA) pipeline** for humanoid manipulation, combining VLM-based instruction following and **imitation learning** with demonstrations collected through VR teleoperation.
- Automated the **URBAN-SIM** asset pipeline with GPT-4o, Grounded DINO, and SAM, supporting **10,000+ interactive 3D assets** for procedural obstacle placement.
- Led **sim-to-real deployment** on COCO robots, evaluating navigation policies across **50+ unseen urban scenarios**.

UCLA, Computational Machine Learning Lab — Mar 2023 – Sep 2024
Research Assistant, Generative Models — *Advisor: Prof. Cho-Jui Hsieh*
- Developed a **training-free fingerprinting method** using layer-wise memorization profiles; studied embedding-space selection for diffusion and GAN models with **ViT and CNN encoders**.

Selected Projects

On-Device VLM Inference Optimization — Oct 2025 – Dec 2025
- Optimized **Qwen3-VL-2B** inference on **Snapdragon 8 Elite**, achieving a **2.05x speedup** over the project baseline.
- Configured **llama.cpp** with importance-matrix-based **Q4_0 quantization**, memory locking (`mlock`), and **CPU affinity** for on-device inference.
- Fine-tuned with **LoRA** for visual question answering, evaluating both task quality and inference performance.

High-Performance C++ Web Server — Mar 2025 – Jun 2025
- Built an asynchronous **C++** web server using **Boost.Asio**, Redis caching, and PostgreSQL connection pooling to support **1,000+ concurrent users**.
- Led a four-person team using TDD; automated CI/CD with **Docker** and **Google Cloud Build**.

Selected Publications

- **Jack He**, Jianxing Zhao, Andrew Bai, Cho-Jui Hsieh. *Layer Choice for Memorization Detection and Fingerprinting for Generative Models.* arXiv.
- Wayne Wu et al. (co-author). *MetaUrban: An Embodied AI Simulation Platform for Urban Micromobility.* **ICLR 2025, Spotlight**.
- Wayne Wu et al. (co-author). *Towards Autonomous Micromobility through Scalable Urban Simulation.* **CVPR 2025, Spotlight**.